

Rohan Gupta

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Looking for roles in AI/ML Research and Engineering

EDUCATION

UNIVERSITY OF SOUTHERN CALIFORNIA

BS Computer Science, Finance Minor

MS Computer Science, emphasis in AI/ML & Robotics

GPA: 3.73/4

Relevant Activities: Undergraduate TA for introductory Artificial Intelligence and Operating Systems courses

Los Angeles, CA

Aug 2020 - May 2024

Jan 2024 to May 2025

RELEVANT EXPERIENCE

SYNOPSYS | Senior Machine Learning Engineer

R&D Machine Learning Intern

July 2025 – Present

June 2024 – Feb 2025

- Owned research and full-stack implementation (~100K LoC across ~160K-line codebase) within a 3-person team of a transformer-based model for intent-based multi-stage chip design optimization that generated recommendations that achieved Pareto-dominant power-timing tradeoffs that **beat baseline expert-designed recommendations** for placement, clock tree synthesis, and routing in **93% of evaluations** on proprietary designs, achieving up to **75% lower TNS, 21% lower WNS, 84% lower NVP, 2% lower power, 40% lower leakage**
- Designed a GPT-inspired decoder model with multi-source embedding composition, EDA feature encodings, and multi-task heads (logits, EDA score prediction, etc.) that generates recommendation tokens for chip designs
- Engineered production-grade ML infrastructure (Docker, Hydra, MLFlow, PyTorch, FSDP/DDP) with custom logging for activation/gradient metrics across **200+ training runs**, and built a full-stack Python/React dashboard for analyzing training dynamics and EDA performance metrics
- Implemented pre-training (**curriculum learning, DPO, multi-objective MSE-based losses**) and online RL finetuning (**GRPO**) using EDA tool calls as environment/reward
- Built inference-time decoding (beam search, greedy, sampling) conditioned on optimization intent (timing, power, leakage) and stage (placement, CTS, routing)
- Deployed inference infrastructure for the chip optimization model via a containerized service on GPU hosts with the EDA environment runtime, using batching and in-memory model caching for low-latency repeated inference; used vLLM and LangChain to serve and orchestrate both post-trained (Qwen3.5-9B) and base LLMs, with the model analytics dashboard launching and querying inference sessions directly
- Built the data collection and featurization pipeline (EDA tools, proprietary design DBs, custom tokenizers) and optimized training via data stores/caches
- Contributed to secondary projects spanning netlist tokenization, circuit foundation models, JAX and CUDA-based speed ups of differentiable optimization in chip placement algorithms, LLM agents for flagship Fusion Compiler product, sparse autoencoder-interpretability pipelines, and post-training using proprietary data

USC VITERBI SCHOOL OF ENGINEERING | AI/ML Research Assistant & Engineer

May 2023 – July 2025

- Won university **research award** to collaborate with Professor Jesse Thomason in the GLAMOR Lab on robotics, computer vision, and NLP research for Vision-and-Language Navigation and Sim2Real
- Collected, trained and fine-tuned language-conditioned policies (ACT, Cross Modal Attention, Diffusion, etc.) on multimodal robot data for mobile manipulators and navigation robots using reinforcement learning for finetuning and supervised learning, specifically behavior-cloning
- Contributed to a heterogeneous multi-robot coordination codebase using ROS1/ROS2 and diverse hardware libraries (Toyota HSR, Trossen, Stretch)
- Worked on top of public repos like TeleMoMa and LeRobot and fixed bugs in them as well as added ROS1 to ROS2 upgrades for them
- Built a sample-efficient evaluation framework for a physical robot using perplexity and distance-to-goal as success metrics
- Researched how contrast sets can be applied to robot experiments to save time and experimenter effort — **co-first author** on both papers for this project that got accepted to a [CoRL 2023 Workshop](#) (Oral, **Top 15%** of papers) and [CoRL 2024](#) (38.2% acceptance rate)
- Built simulation data collection and policy training pipelines using MetaWorld for an efficient policy evaluation project that got accepted to [CoRL 2025](#)
- Led project “RobotFleet” — a heterogeneous multi-robot coordination using natural language LLMs and multimodal policies. Paper accepted to RSS 2025 Multi-Robot Systems Workshop and GitHub repository [open-sourced](#) and being maintained

INTERNSHIPS AND OTHER PROJECTS | Co-founder, student developer, intern

- USC Investment Office** Jun – Dec 2022
Focused on software, automation, and time-series financial and data analysis projects to support running **\$8B+** endowment.
- StreetFins** Oct 2017 – Nov 2022
Built and grew **team of 60+ students** for a financial education startup. Created the business plan, coded the website, and reached out successfully to high schools across **16 states**. Hosted the “Finance Simplified” podcast which has over **70k** listens with notable guests from industry, academia, and government. Won **paid contracts** with 2 larger companies to produce educational content.

SKILLS

- Languages:** Python, C/C++, Java, JavaScript, TypeScript, Angular, React, PHP, SQL
- ML/Data:** PyTorch, JAX, CUDA, Hydra, WandB, MLFlow, Vector DBs, LangChain, Stable Baselines, FSDP/DDP, GRPO/PPO/DPO, NumPy, Pandas, HuggingFace
- Tools:** Docker, Linux, AWS, ROS1/2, Git, OpenCV